

11 – Linux

A free and open source Operating System

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OUTLINE

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3) Commands

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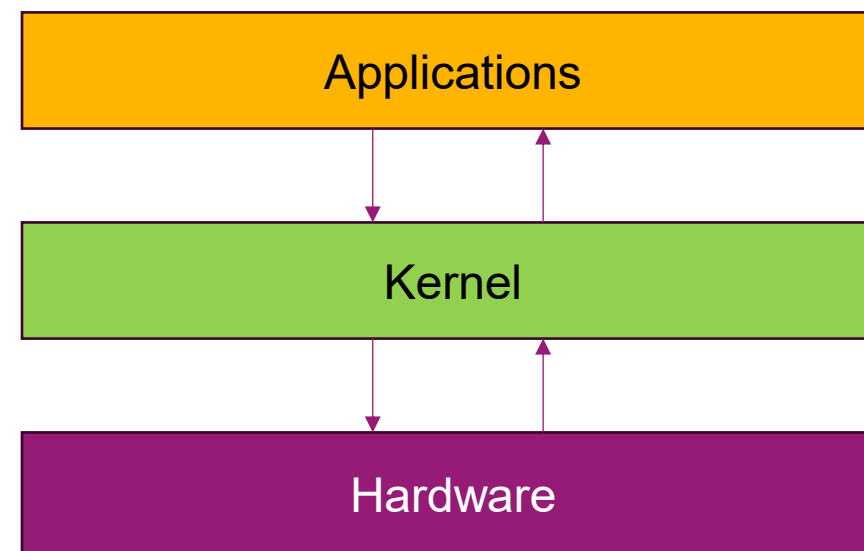
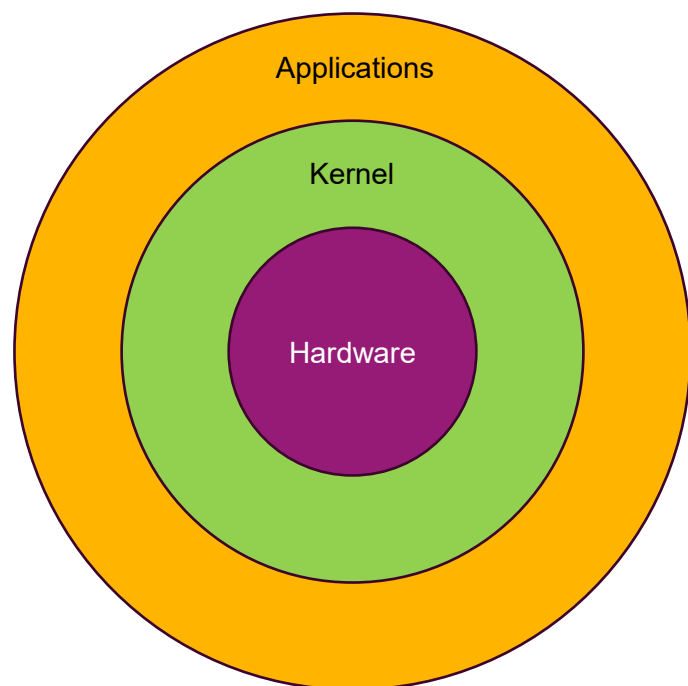
3.11) System Information

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1) LINUX

- **Linux** is a **Unix-like Operating System** which is free and open-source
- All Linuxes share the Linux **kernel** with the **GNU** utilities
- A **distribution** is a certain Linux kernel with **applications** and **services**

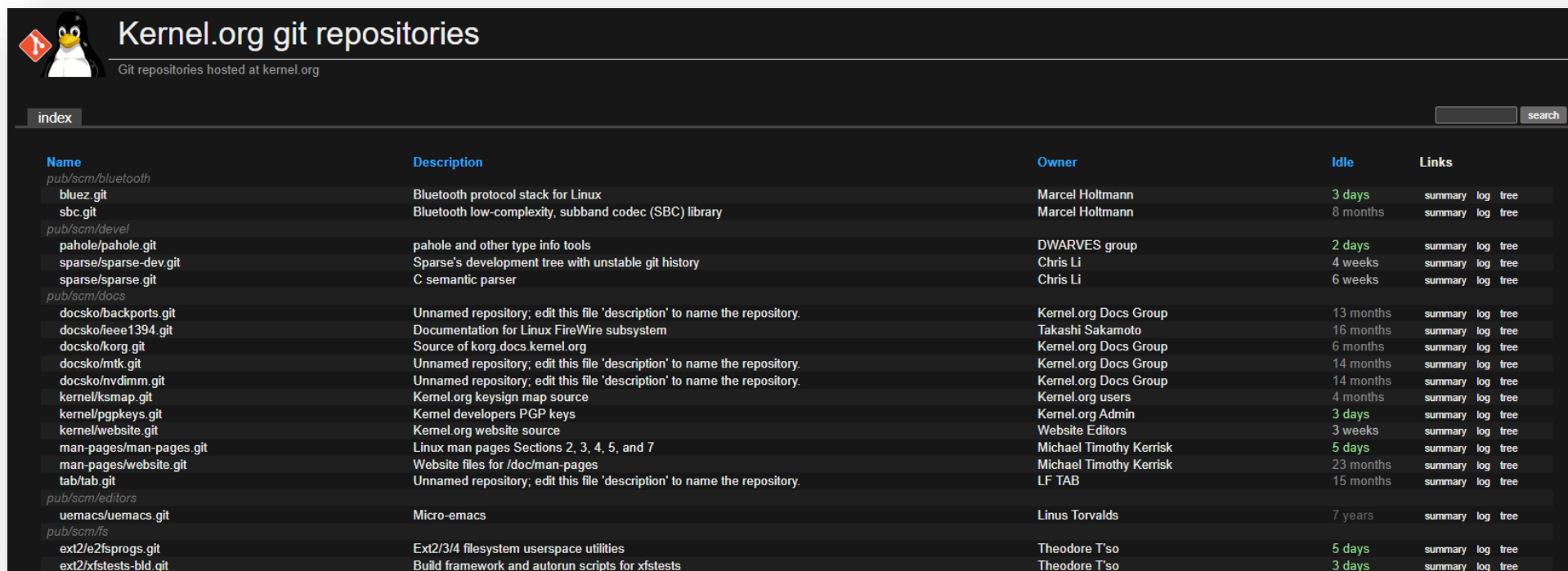


1) LINUX

- **The Linux kernel** has been developed by **Linus TORVALDS** and put in the public domain in 1991 (version 0.01)
- **The GNU** (Richard STALLMAN) and **Free Software Foundation** made Linux a free and open source project: GNU/Linux

1) LINUX

- **The Linux kernel** is a modular software in which you can choose which modules you want to use
- It means you can build and compile **your own version** of the kernel with a list of modules that target your physical (embedded) device
- You can have a look at the progression of the linux kernel on <https://git.kernel.org/>



The screenshot shows the 'Kernel.org git repositories' index page. It features a table with columns for Name, Description, Owner, Idle, and Links. The table lists various repositories such as 'pub/scm/bluetooth', 'pub/scm/devel', 'pub/scm/docs', 'pub/scm/editors', and 'pub/scm/fs'. Each repository entry includes its name, a brief description, the owner's name, the time it has been idle, and links to summary, log, and tree views.

Name	Description	Owner	Idle	Links
<i>pub/scm/bluetooth</i>				
bluez.git	Bluetooth protocol stack for Linux	Marcel Holtmann	3 days	summary log tree
sbcc.git	Bluetooth low-complexity, subband codec (SBC) library	Marcel Holtmann	8 months	summary log tree
<i>pub/scm/devel</i>				
pahole/pahole.git	pahole and other type info tools	DWARVES group	2 days	summary log tree
sparse/sparse-dev.git	Sparse's development tree with unstable git history	Chris Li	4 weeks	summary log tree
sparse/sparse.git	C semantic parser	Chris Li	6 weeks	summary log tree
<i>pub/scm/docs</i>				
docsko/backports.git	Unnamed repository; edit this file 'description' to name the repository.	Kernel.org Docs Group	13 months	summary log tree
docsko/ieee1394.git	Documentation for Linux FireWire subsystem	Takashi Sakamoto	16 months	summary log tree
docsko/korg.git	Source of korg docs.kernel.org	Kernel.org Docs Group	6 months	summary log tree
docsko/mtk.git	Unnamed repository; edit this file 'description' to name the repository.	Kernel.org Docs Group	14 months	summary log tree
docsko/nvdim.git	Unnamed repository; edit this file 'description' to name the repository.	Kernel.org Docs Group	14 months	summary log tree
kernel/ksmap.git	Kernel.org keysign map source	Kernel.org users	4 months	summary log tree
kernel/pgpkeys.git	Kernel developers PGP keys	Kernel.org Admin	3 days	summary log tree
kernel/website.git	Kernel.org website source	Website Editors	3 weeks	summary log tree
man-pages/man-pages.git	Linux man pages Sections 2, 3, 4, 5, and 7	Michael Timothy Kerrisk	5 days	summary log tree
man-pages/website.git	Website files for /doc/man-pages	Michael Timothy Kerrisk	23 months	summary log tree
tab/tab.git	Unnamed repository; edit this file 'description' to name the repository.	LF TAB	15 months	summary log tree
<i>pub/scm/editors</i>				
uemacs/uemacs.git	Micro-emacs	Linus Torvalds	7 years	summary log tree
<i>pub/scm/fs</i>				
ext2/e2fsprogs.git	Ext2/3/4 filesystem userspace utilities	Theodore T'so	5 days	summary log tree
ext2/xfstests-bld.git	Build framework and autorun scripts for xfstests	Theodore T'so	3 days	summary log tree

1) LINUX

- **Each Linux kernel** has a version number:
 - Linux 6.10.x – Production version
 - Linux 6.11.x – Experimental (development) version
 - Linux 6.12.x – Production version

KERNELVERSION=\$(VERSION).\$(PATCHLEVEL).\$(SUBLEVEL)\$(EXTRAVERSION)

- The official documentation is available on <https://www.kernel.org/doc/html/latest/>



The Linux Kernel

6.18.0-rc6

Quick search

Contents

Development process
Submitting patches
Code of conduct
Maintainer handbook
All development-process docs
Core API
Driver APIs

The Linux Kernel documentation

This is the top level of the kernel's documentation tree. Kernel documentation, like the kernel itself, is very much a work in progress; that is especially true as we work to integrate our many scattered documents into a coherent whole. Please note that improvements to the documentation are welcome; join the linux-doc list at vger.kernel.org if you want to help out.

English

Working with the development community

The essential guides for interacting with the kernel's development community and getting your work upstream.

- [Development process](#)
- [Submitting patches](#)
- [Code of conduct](#)
- [Maintainer handbook](#)
- [All development-process docs](#)

Internal API manuals

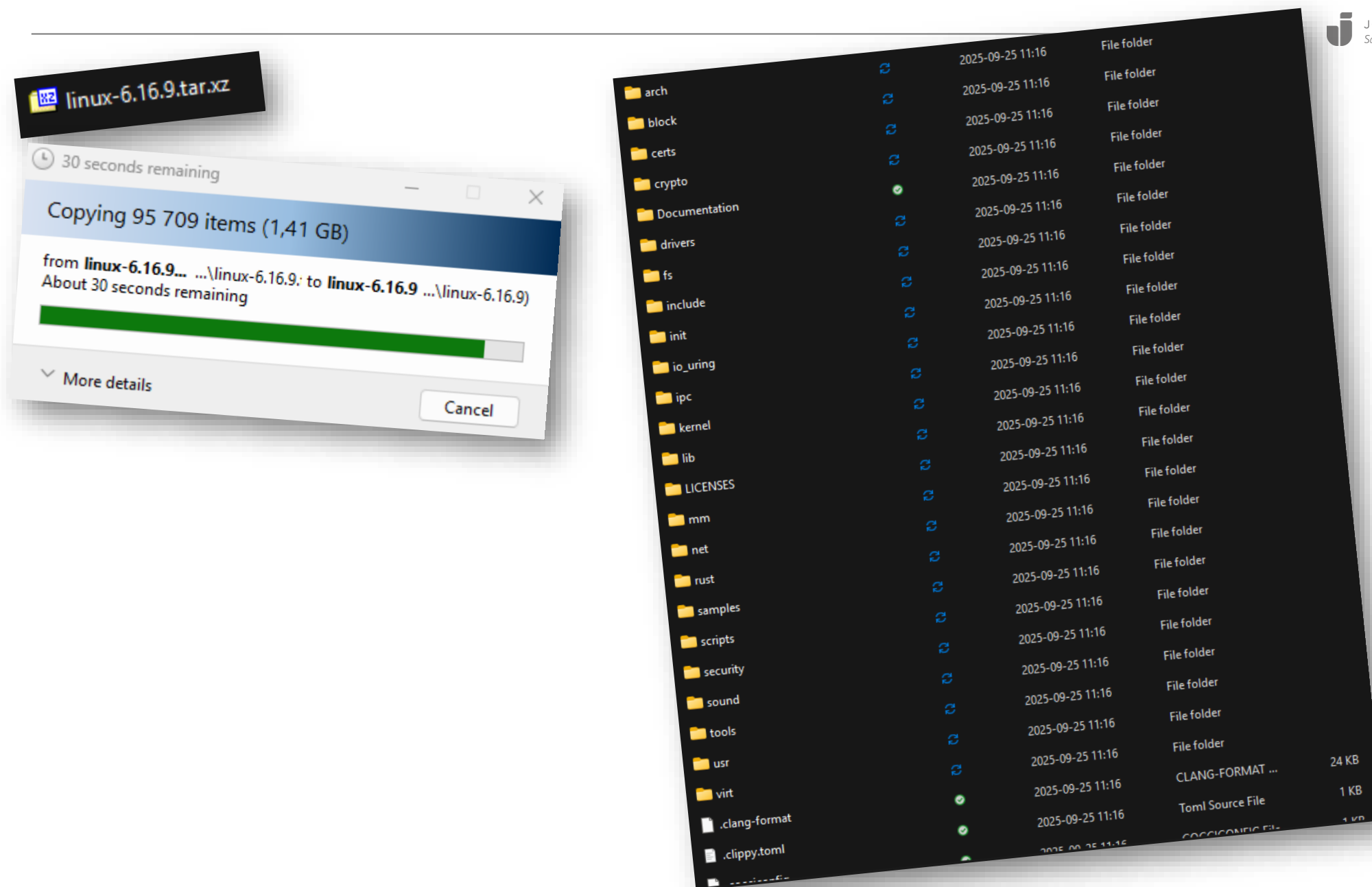
Manuals for use by developers working to interface with the rest of the kernel.

- [Core API](#)
- [Driver APIs](#)
- [Subsystems](#)
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arc	2025-09-25 11:16	File folder	
arm	2025-09-25 11:16	File folder	
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audit.h		2025-09-25 11:16	C Source File	66 KB
		2025-09-25 11:16	C Header Source F...	11 KB

kernel/

 fork.c

kernel/

```
1 // SPDX-License-Identifier: GPL-2.0-only
2 /*
3  *   linux/kernel/fork.c
4  *
5  *   Copyright (C) 1991, 1992   Linus Torvalds
6  */
7
8 /*
9  *   'fork.c' contains the help-routines for the 'fork' system call
10  *   (see also entry.S and others).
11  *   Fork is rather simple, once you get the hang of it, but the memory
12  *   management can be a bitch. See 'mm/memory.c': 'copy_page_range()'
13  */
14
15 #include <linux/anon_inodes.h>
16 #include <linux/slab.h>
17 #include <linux/sched/autogroup.h>
18 #include <linux/sched/mm.h>
19 #include <linux/sched/user.h>
20 #include <linux/sched/numa_balancing.h>
21 #include <linux/sched/stat.h>
22 #include <linux/sched/task.h>
23 #include <linux/sched/task_stack.h>
24 #include <linux/sched/cputime.h>
25 #include <linux/sched/ext.h>
26 #include <linux/seq_file.h>
```

 fork.c

kernel/

```
164 int nr_processes(void)
165 {
166     int cpu;
167     int total = 0;
168
169     for_each_possible_cpu(cpu)
170         total += per_cpu(process_counts, cpu);
171
172     return total;
173 }
```

```

259 static int memcg_charge_kernel_stack(struct vm_struct *vm)
260 {
261     int i;
262     int ret;
263     int nr_charged = 0;
264
265     BUG_ON(vm->nr_pages != THREAD_SIZE / PAGE_SIZE);
266
267     for (i = 0; i < THREAD_SIZE / PAGE_SIZE; i++) {
268         ret = memcg_kmem_charge_page(vm->pages[i], GFP_KERNEL, 0);
269         if (ret)
270             goto err;
271         nr_charged++;
272     }
273     return 0;
274 err:
275     for (i = 0; i < nr_charged; i++)
276         memcg_kmem_uncharge_page(vm->pages[i], 0);
277     return ret;
278 }

```

```
1905  /*
1906  * This creates a new process as a copy of the old one,
1907  * but does not actually start it yet.
1908  *
1909  * It copies the registers, and all the appropriate
1910  * parts of the process environment (as per the clone
1911  * flags). The actual kick-off is left to the caller.
1912  */
1913  latent_entropy struct task_struct *copy_process(
1914      struct pid *pid,
1915      int trace,
1916      int node,
1917      struct kernel_clone_args *args)
1918  {
1919      int pidfd = -1, retval;
1920      struct task_struct *p;
1921      struct multiprocess_signals delayed;
1922      struct file *pidfile = NULL;
1923      const u64 clone_flags = args->flags;
1924      struct nsproxy *nsp = current->nsproxy;
1925
1926      /*
1927       * Don't allow sharing the root directory with processes in a different
1928       * namespace
1929       */
1930      if ((clone_flags & (CLONE_NEWNS|CLONE_FS)) == (CLONE_NEWNS|CLONE_FS))
1931          return ERR_PTR(-EINVAL);
```

2308
2309
2310

```
p->start_time = ktime_get_ns();  
p->start_boottime = ktime_get_boottime_ns();
```

```
2317     /* CLONE_PARENT re-uses the old parent */  
2318     if (clone_flags & (CLONE_PARENT|CLONE_THREAD)) {  
2319         p->real_parent = current->real_parent;  
2320         p->parent_exec_id = current->parent_exec_id;  
2321         if (clone_flags & CLONE_THREAD)  
2322             p->exit_signal = -1;  
2323         else  
2324             p->exit_signal = current->group_leader->exit_signal;  
2325     } else {  
2326         p->real_parent = current;  
2327         p->parent_exec_id = current->self_exec_id;  
2328         p->exit_signal = args->exit_signal;  
2329     }  
2330  
2331     klp_copy_process(p);  
2332  
2333     sched_core_fork(p);  
2334  
2335     spin_lock(&current->sighand->siglock);  
2336  
2337     rv_task_fork(p);  
2338  
2339     rseq_fork(p, clone_flags);
```

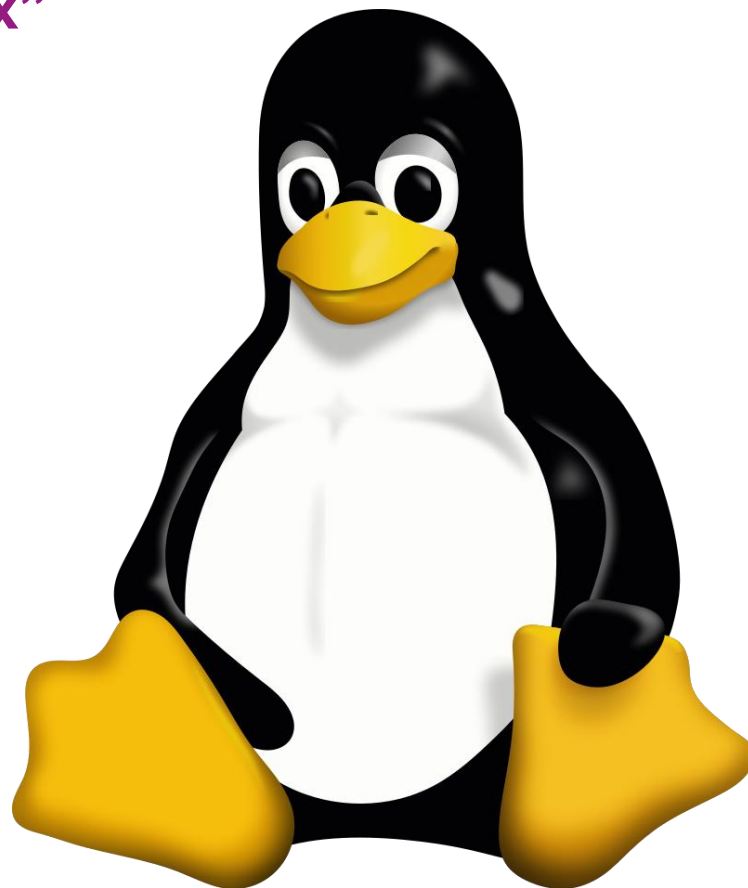


```
2402     total_forks++;
2403     hlist_del_init(&delayed.node);
2404     spin_unlock(&current->sigband->siglock);
2405     syscall_tracepoint_update(p);
2406     write_unlock_irq(&tasklist_lock);
2407
2408     if (pidfile)
2409         fd_install(pidfd, pidfile);
2410
2411     proc_fork_connector(p);
2412     sched_post_fork(p);
2413     cgroup_post_fork(p, args);
2414     perf_event_fork(p);
2415
2416     trace_task_newtask(p, clone_flags);
2417     uprobe_copy_process(p, clone_flags);
2418     user_events_fork(p, clone_flags);
2419
2420     copy_oom_score_adj(clone_flags, p);
2421
2422     return p;
```

```
2507 struct task_struct * __init fork_idle(int cpu)
2508 {
2509     struct task_struct *task;
2510     struct kernel_clone_args args = {
2511         .flags      = CLONE_VM,
2512         .fn         = &idle_dummy,
2513         .fn_arg     = NULL,
2514         .kthread    = 1,
2515         .idle       = 1,
2516     };
2517
2518     task = copy_process(&init_struct_pid, 0, cpu_to_node(cpu), &args);
2519     if (!IS_ERR(task)) {
2520         init_idle_pids(task);
2521         init_idle(task, cpu);
2522     }
2523
2524     return task;
2525 }
```

1) LINUX

- Linux has a mascot: **“Tux”**



1) LINUX

- Linux has a mascot: **“Tux”**

Tux variants



Vectorized from the pixel image source



Gloss look Tux



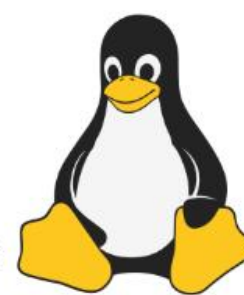
High quality 3d-look variant



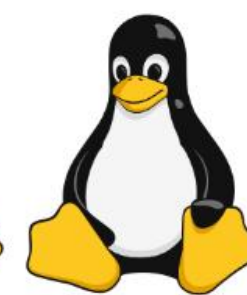
Modern flat look



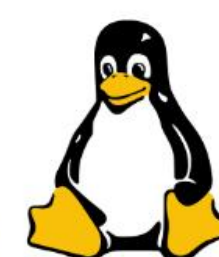
Modern flat look in monochrome



Classic flat look: Emblem



Classic flat look: 3D



Simple vector variant



Tux Crystal 1st revision



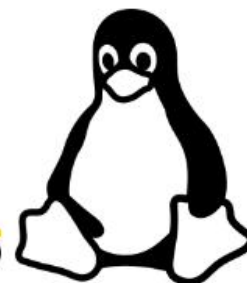
Tux Crystal 2nd revision



Tux from TuxGuitar



Slackware version of Tux



A simplified flat Tux



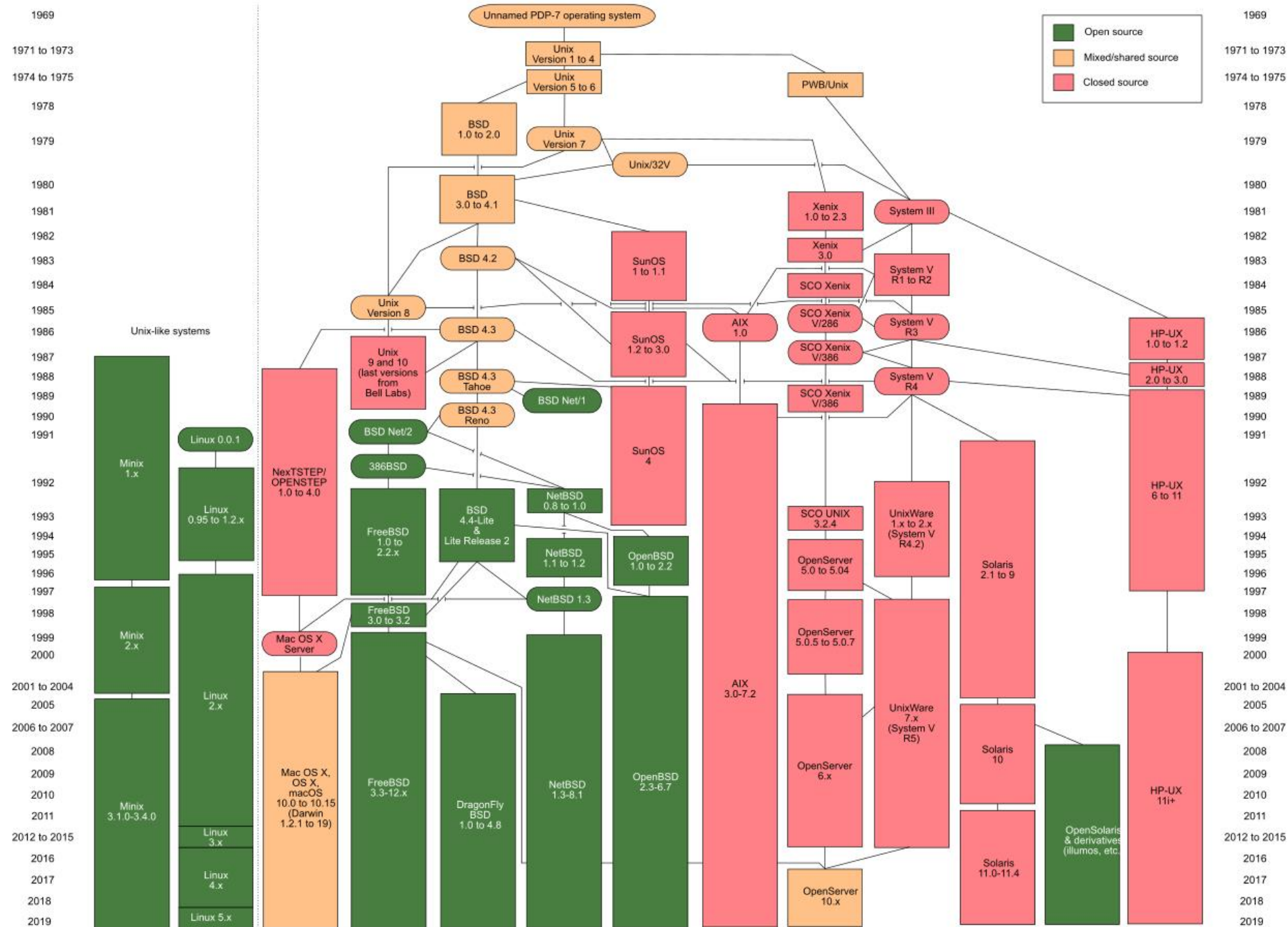
The Other Linux Logo



The Other Linux Logo (sitting version)



PaX version of Tux



1) LINUX

- Many distributions:



Page Hit Ranking		
Data span:		
	Year 2023	Go
Rank	Distribution	HPD*
1	MX Linux	2668
2	Mint	2097
3	EndeavourOS	1942
4	Debian	1360
5	Manjaro	1335
6	Ubuntu	1079
7	Fedora	1017
8	Pop!_OS	1007
9	openSUSE	755
10	Zorin	705
11	Lite	627
12	Garuda	612
13	elementary	539
14	KDE neon	533
15	antiX	496
16	Nobara	477
17	Mageia	403
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3CX

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Ubuntu 25.04 may offer level 3 (O3) compiler optimizations	
Tumbleweed gains major updates for LibreSSL, Qt, GTK4, and Network Manager	
Raspberry Pi foundation launches new line of SSDs	
AlmaLinux launches the Kitten development branch	
Ubuntu turns 20 years old	
UBports adding VoLTE support	
Debian starts adoption of Perl 5.40	
KDE neon upgrades its base	
New improvements in Haiku	
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2024-11-04

[DistroWatch Weekly, Issue 1095](#)

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- [News: Raspberry Pi computers gain official SSD storage options, openSUSE Tumbleweed introduces several major upgrades, Ubuntu testing compiler optimizations, Mint partnering with Framework](#)
- [Questions and answers: Get a list of installed applications](#)
- [Released last week: Fedora 41, Raspberry Pi OS 2024-10-22, BackBox Linux 9, Linux Lite 7.2, TrueNAS 24.10.0, Br OS 24.10](#)
- [Torrent corner: ArcoLinux, Fedora, Linux Lite, Raspberry Pi OS, Tails, TUXEDO OS](#)
- [Upcoming releases: FreeBSD 14.2 BETA2](#)
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Pisi Linux

2024-11-01

[Distribution Release: Pisi Linux 2.4](#)

Erkan Işık has announced the release of [Pisi Linux 2.4](#), code name "Karagül", the latest stable version from the project that builds an independently-developed Linux distribution for the desktop. It uses the Pisi package management tool, also developed in-house. The new version updates the KDE Plasma desktop to version 6.2 and the Linux kernel to version 6.6.56. "Inspired by the rare flowers of Anatolia, Pisi GNU/Linux delivers the 2.4 version 'Karagül' to its users. This version is designed to offer users a modern, fast and flexible experience. Pisi GNU/Linux 2.4 'Karagül' is updated with the latest version of the KDE desktop environment, KDE Plasma 6. This version promises its users a more contemporary, rich and efficient desktop experience by bringing together the Wayland graphics platform and the Qt6 application framework. With its features that emphasize performance and stability, a more efficient working environment awaits on your computer." Read the complete [release announcement](#) (in Turkish) for more information. Besides Turkish, the distribution supports English and several European languages. [Download \(MD5\)](#) link: [Pisi-Linux-2.4-KDE6-x86_64.iso](#) (3,159MB, [pkglist](#)). A separate installation-only "Minimal" image is also available: [Pisi-Linux-2.4-Minimal-x86_64.iso](#) (1,680MB).

[Rate this project](#)



LINUX LITE

2024-10-31

[Distribution Release: Linux Lite 7.2](#)

Jerry Bezencon has announced the release of [Linux Lite 7.2](#), an update to the project's Ubuntu-based distribution featuring the Xfce desktop. The new release includes a new update manager, a new theme manager, and the website now offers a chatbot intended to offer technical support for Linux Lite's community. "Linux Lite is leading the way in which AI is complementing the support portion of our operating system. We're extremely happy with the way that it's turned out and a huge thank you again to Huzaifa Qureshi for his efforts on this. Click [here](#) to take a look. Lite Updates - Re-written in Yad. Lite Updates has been re-written in Yad as Yad provides more flexibility than Zenity and will help us to expand and add more options to it in the future. Lite Theme Manager: Brand new to the Linux Lite family of Applications is Lite Theme Manager. Lite Theme Manager takes the guess work out of adding themes to your desktop, icons and mouse cursors with an easy point-and-click solution. Thank you and credit goes to Huzaifa Qureshi for writing this application." Additional information and screenshots can be found in the

3CX

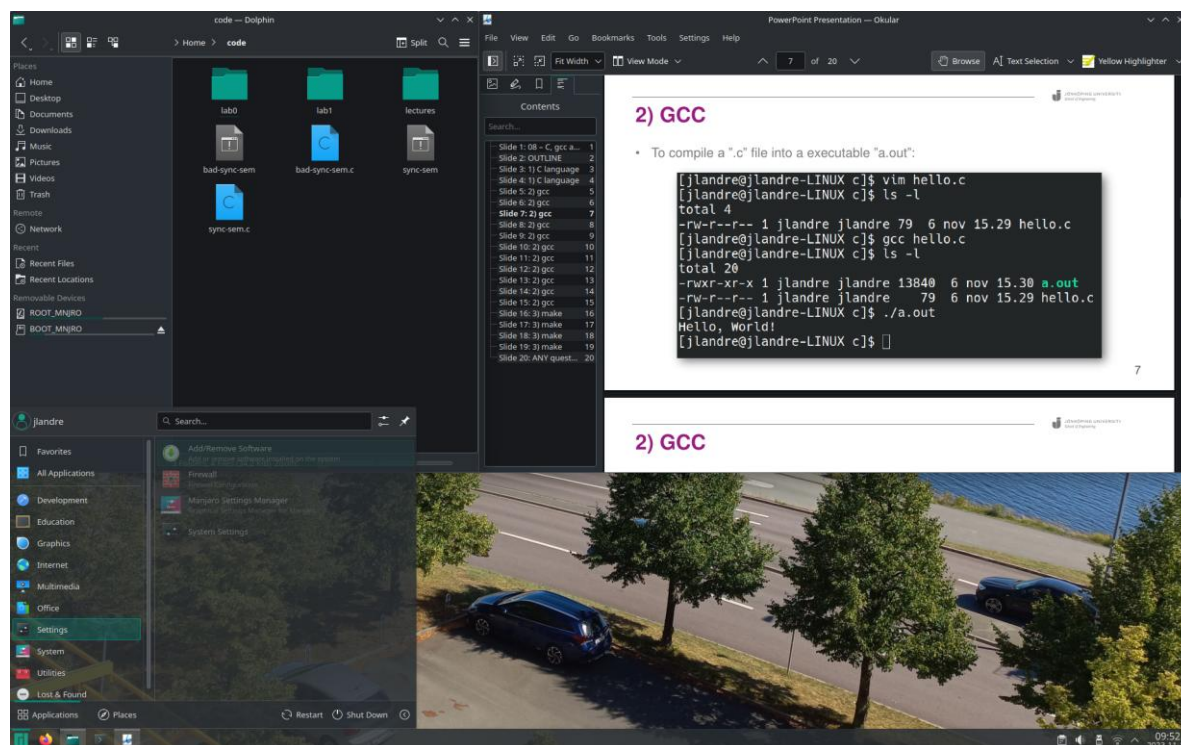
Free and easy-to-use

Page Hit Ranking		
Data span:		
Year 2023 Go		
Rank	Distribution	HPD*
1	MX Linux	2668
2	Mint	2097
3	EndeavourOS	1942
4	Debian	1360
5	Manjaro	1335
6	Ubuntu	1079
7	Fedora	1017
8	Pop!_OS	1007
9	openSUSE	755
10	Zorin	705
11	Lite	627
12	Garuda	612
13	elementary	539
14	KDE neon	533
15	antiX	496
16	Nobara	477
17	Mageia	403
18	Kali	400
19	Puppy	385
20	SparkyLinux	381
21	AlmaLinux	371
22	PCLinuxOS	369
23	Vanilla	366
24	FreeBSD	363
25	ArcoLinux	359
26	NixOS	341
27	Alpine	334
28	EasyOS	330
29	Q4OS	311
30	CachyOS	301
31	Peppermint	285

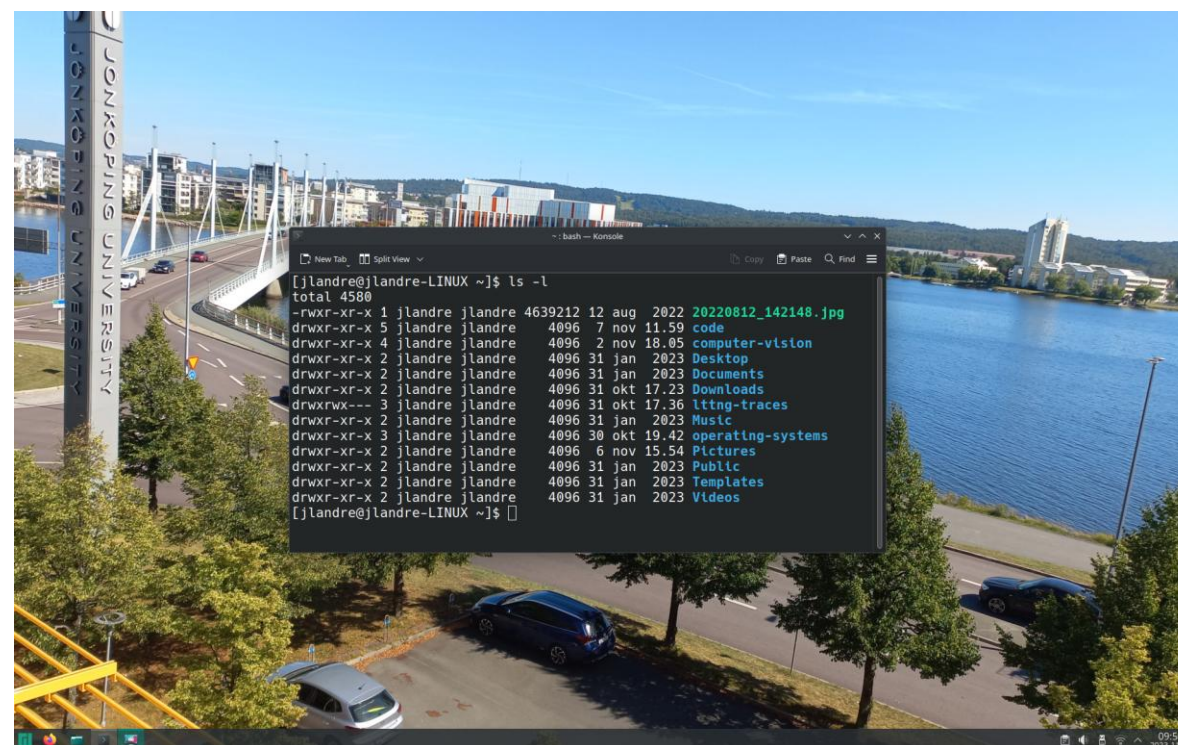
2) SHELLS

- A **shell** is an interface to access the Linux system environment
- There are two different shells on Linux:

The **GUI**



The **Terminal**



2) SHELLS

- Both the **GUI** and the **Terminal** are useful but their usage is different
- The GUI is good at “simple” tasks with only a few graphical elements are involved in the operation
- The Terminal is good at doing “massive” work: rename the files in a directory, move all the JPEG files from a directory to another...

3) COMMANDS

- In the **terminal**, you can type in many **commands** and install even more
- Commands are related to:
 - Getting help
 - Users, groups, and users/groups management
 - System update and package management
 - The filesystem tree, paths, directories, files and permissions
 - Tasks: Processes/Threads
 - Memory
 - Disk
 - Networking
 - System information
 - Other commands
 - ...

3.1) GETTING HELP

- The most important command in Linux is **man**
- It gives information and help about commands, C functions, system tools...

```
os2024@os2024-virtualbox: ~ 103x26
APT(8)
NAME
    apt - command-line interface
SYNOPSIS
    apt [-h] [-o=config_string] [-c=config_file] [-t=target_release] [-a=architecture] {list |
    search | show | update | install pkg [{=pkg_version_number | /target_release}]... |
    remove pkg... | upgrade | full-upgrade | edit-sources | {-v | --version} |
    {-h | --help}}
DESCRIPTION
    apt provides a high-level commandline interface for the package management system. It is
    intended as an end user interface and enables some options better suited for interactive
    usage by default compared to more specialized APT tools like apt-get(8) and apt-cache(8).

    Much like apt itself, its manpage is intended as an end user interface and as such only
    mentions the most used commands and options partly to not duplicate information in multiple
    places and partly to avoid overwhelming readers with a cornucopia of options and details.

    update (apt-get(8))
    update is used to download package information from all configured sources. Other
    commands operate on this data to e.g. perform package upgrades or search in and display
    details about all packages available for installation.
Manual page apt(8) line 1 (press h for help or q to quit)
```

os2024@os2024-virtualbox:~\$ man apt

3.1) GETTING HELP

```
os2024@os2024-virtualbox:~$ man printf
```

```
PRINTF(1)                                User Commands                                PRINTF(1)

NAME
    printf - format and print data

SYNOPSIS
    printf FORMAT [ARGUMENT]...
    printf OPTION

DESCRIPTION
    Print ARGUMENT(s) according to FORMAT, or execute according to OPTION:

    --help display this help and exit

    --version
        output version information and exit

    FORMAT controls the output as in C printf.  Interpreted sequences are:

    \"      double quote
    \\      backslash
    \a      alert (BEL)

Manual page printf(1) line 1 (press h for help or q to quit)
```

3.1) GETTING HELP

```
os2024@os2024-virtualbox:~$ man pthread_create
```

```
pthread_create(3)                                Library Functions Manual                                pthread_create(3)

NAME
  pthread_create - create a new thread

LIBRARY
  POSIX threads library (libpthread, -lpthread)

SYNOPSIS
  #include <pthread.h>

  int pthread_create(pthread_t *restrict thread,
                     const pthread_attr_t *restrict attr,
                     void *(*start_routine)(void *),
                     void *restrict arg);

DESCRIPTION
  The pthread_create() function starts a new thread in the calling process. The new thread starts execution by invoking start_routine(); arg is passed as the sole argument of start_routine().

  The new thread terminates in one of the following ways:

  • It calls pthread_exit(3), specifying an exit status value that is available to another thread in the same process that calls pthread_join(3).

Manual page pthread_create(3) line 1 (press h for help or q to quit)
```

3.2) USERS/GROUPS MANAGEMENT

```
$ adduser jerome
```

```
$ groupadd groupname
```

```
$ chown owner:group name
```

```
$ chgrp groupname filename.txt
```

```
$ sudo usermod -a -G groupname jerome
```


3.2) USERS/GROUPS MANAGEMENT

```
os2024@os2024-virtualbox:~$ sudo adduser jlandre
info: Adding user `jlandre' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `jlandre' (1003) ...
info: Adding new user `jlandre' (1003) with group `jlandre (1003)' ...
info: Creating home directory `/home/jlandre' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jlandre
Enter the new value, or press ENTER for the default
    Full Name []: Jerome Landre
    Room Number []: E3418
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `jlandre' to supplemental / extra groups `users' ...
info: Adding user `jlandre' to group `users' ...
os2024@os2024-virtualbox:~$
```

3.2) USERS/GROUPS MANAGEMENT

```
os2024@os2024-virtualbox:~$ ls -l /home/
total 8
drwxr-x---  2 jlandre jlandre 4096 nov 12 08:01 jlandre
drwxr-x--- 18 os2024  os2024 4096 nov 12 08:10 os2024
os2024@os2024-virtualbox:~$
```

```
os2024@os2024-virtualbox:~$ sudo addgroup teachers
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `teachers' (GID 1001) ...
os2024@os2024-virtualbox:~$
```

3.2) USERS/GROUPS MANAGEMENT

```
os2024@os2024-virtualbox:~$ groups jlandre
jlandre : jlandre users
os2024@os2024-virtualbox:~$
```

```
os2024@os2024-virtualbox:~$ sudo usermod -aG teachers jlandre
os2024@os2024-virtualbox:~$ groups jlandre
jlandre : jlandre users teachers
os2024@os2024-virtualbox:~$
```

3.2) USERS/GROUPS MANAGEMENT

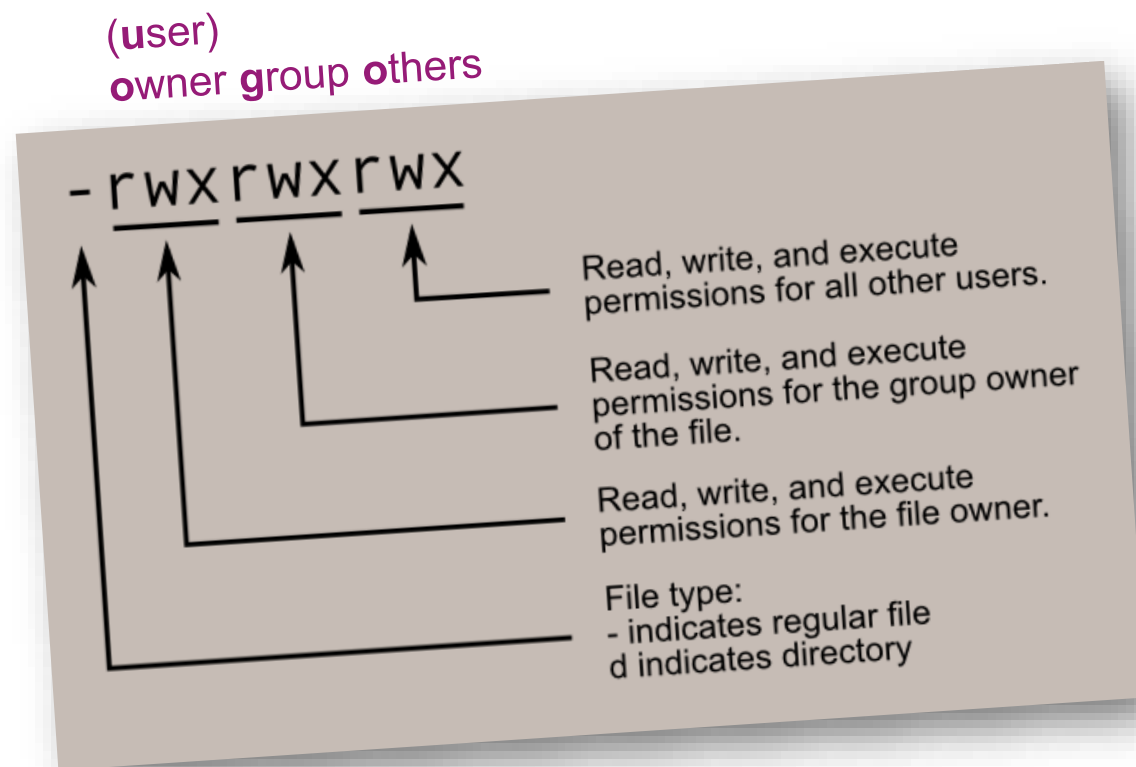
```
os2024@os2024-virtualbox:~$ su - jlandre
Password:
jlandre@os2024-virtualbox:~$ ls -l
total 0
jlandre@os2024-virtualbox:~$ pwd
/home/jlandre
jlandre@os2024-virtualbox:~$ touch myfile.txt
jlandre@os2024-virtualbox:~$ echo "Hej!" > myfile.txt
jlandre@os2024-virtualbox:~$ echo "Hi!" >> myfile.txt
jlandre@os2024-virtualbox:~$ echo "Bonjour!" >> myfile.txt
jlandre@os2024-virtualbox:~$ more myfile.txt
Hej!
Hi!
Bonjour!
jlandre@os2024-virtualbox:~$ ls -l
total 4
-rw-rw-r-- 1 jlandre jlandre 18 nov 12 08:41 myfile.txt
jlandre@os2024-virtualbox:~$
```

3.2) USERS/GROUPS MANAGEMENT

```
jlandre@os2024-virtualbox:~$ chown jlandre:teachers myfile.txt
jlandre@os2024-virtualbox:~$ ls -l
total 4
-rw-rw-r-- 1 jlandre teachers 18 nov 12 08:41 myfile.txt
jlandre@os2024-virtualbox:~$ chmod 660 myfile.txt
jlandre@os2024-virtualbox:~$ ls -l
total 4
-rw-rw---- 1 jlandre teachers 18 nov 12 08:41 myfile.txt
jlandre@os2024-virtualbox:~$
```

3.3) PERMISSIONS/PRIVILEGES/RIGHTS

- Linux uses a system of rights on files and directories



Owner (user who owns the file)
Group (group that owns the file)
Others (everyone else)

4 **r** = Read
2 **w** = Write
1 **x** = Execute

3.3) PERMISSIONS/PRIVILEGES/RIGHTS

4214-14-- = 754

-rwxr-xr--

Owner	Group	Others
rwx	r-x	r--

Owner: read, write, execute

Group: read, execute

Others: read only

3.3) PERMISSIONS/PRIVILEGES/RIGHTS

Permission	File Meaning	Directory Meaning
r	Can view file contents	Can list directory contents
w	Can modify file	Can create/delete files inside
x	Can run file as a program	Can enter (cd) into directory

3.4) THE FILESYSTEM TREE

- Linux uses a TREE to organize files and folders
- The root of the tree is /
- To define a path to any element in the tree, you have TWO possibilities:
 - ABSOLUTE path
 - RELATIVE path
- Folders and files use PERMISSIONS to define the rights of access

```

~ : bash — Konsole
New Tab Split View
Copy Paste Find

[jlandre@jlandre-LINUX ~]$ ls -l /
total 64
lrwxrwxrwx  1 root root    7  3 jan  2023 bin -> usr/bin
drwxr-xr-x  4 root root 16384  1 jan  1970 boot
drwxr-xr-x 16 root root  3920 20 nov  08.32 dev
drwxr-xr-x 82 root root  4096 17 nov 16.15 etc
drwxr-xr-x  3 root root  4096 31 jan  2023 home
lrwxrwxrwx  1 root root    7  3 jan  2023 lib -> usr/lib
drwx----- 2 root root 16384 17 feb  2023 lost+found
drwxr-xr-x  2 root root  4096  3 jan  2023 mnt
drwxr-xr-x  3 root root  4096 31 jan  2023 opt
dr-xr-xr-x 249 root root    0  1 jan  1970 proc
drwxr-xr-x  5 root root  4096 31 okt 17.49 root
drwxr-xr-x 29 root root   700 17 nov 16.15 run
lrwxrwxrwx  1 root root    7  3 jan  2023/sbin -> usr/bin
drwxr-xr-x  4 root root  4096 13 feb  2023 srv
dr-xr-xr-x 12 root root    0  1 jan  1970 sys
drwxrwxrwt 91 root root  1940 20 nov 08.34 tmp
drwxr-xr-x  8 root root  4096  7 nov 13.18 usr
drwxr-xr-x 12 root root  4096  7 nov 14.39 var
[jlandre@jlandre-LINUX ~]$

```

3.5) DISK AND FILES MANAGEMENT

\$ pwd ### print working directory

\$ ls -l ### or ls -la

\$ touch newfile.txt

\$ mkdir newdirectory/

\$ cd newdirectory/

\$ cd ..

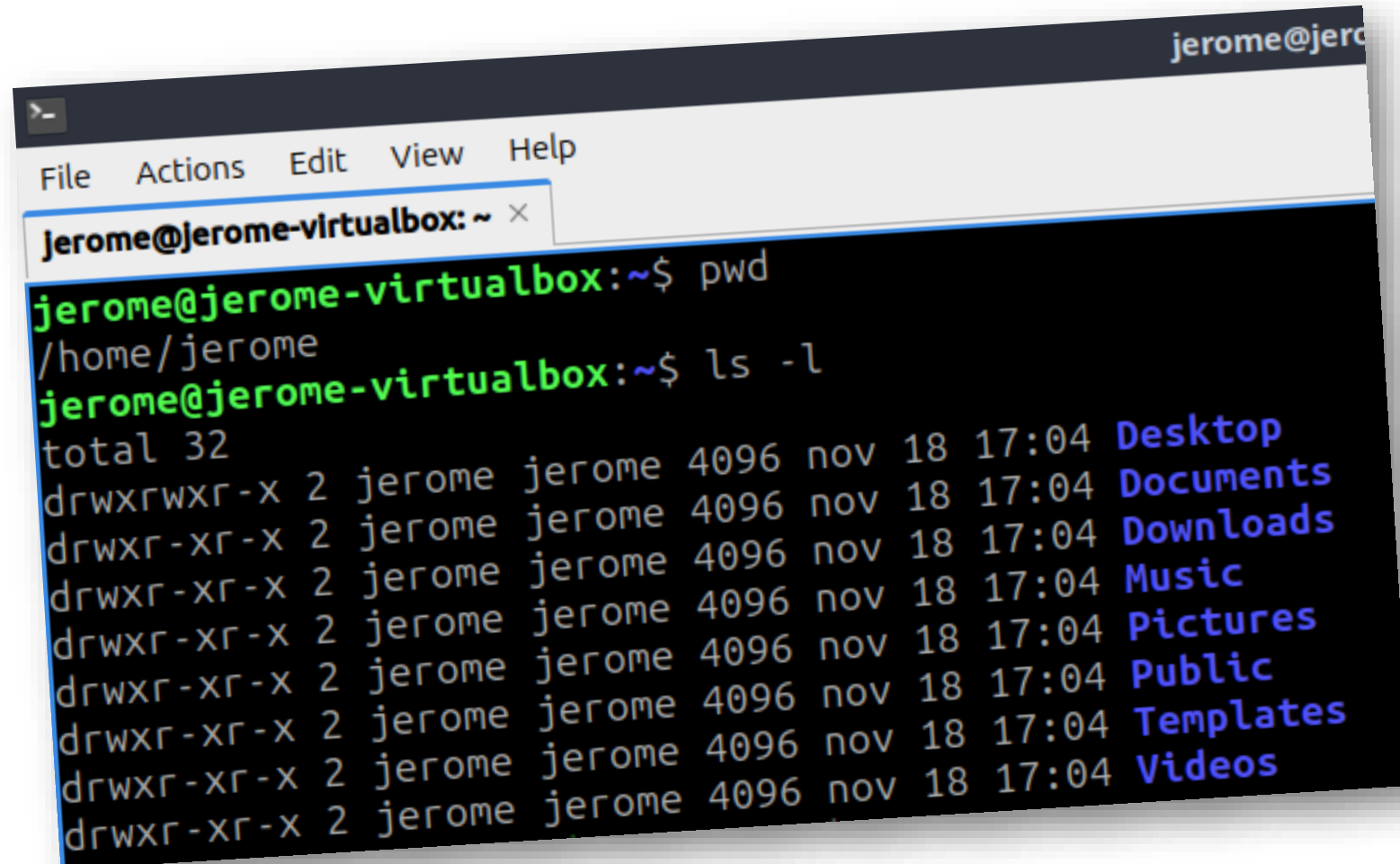
\$ mv oldname newname

\$ cp -Rf sourcename destname

\$ rm filetodelete.txt

\$ rmdir directorytodelete/

3.5) DISK AND FILES MANAGEMENT



The image shows a terminal window titled "jerome@jerome-virtualbox: ~". The window has a menu bar with "File", "Actions", "Edit", "View", and "Help". The terminal content shows the user running the `pwd` command, which returns `/home/jerome`. Then, the user runs `ls -l`, which lists the contents of the home directory. The output shows a total size of 32 and a list of directories with their permissions, owner, group, size, date, and time.

```
jerome@jerome-virtualbox: ~$ pwd
/home/jerome
jerome@jerome-virtualbox: ~$ ls -l
total 32
drwxrwxr-x 2 jerome jerome 4096 nov 18 17:04 Desktop
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Documents
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Downloads
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Music
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Pictures
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Public
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Templates
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Videos
```

```
jerome@jerome-virtualbox:~$ touch myfile.txt
jerome@jerome-virtualbox:~$ ls -la
total 72
drwxr-x--- 13 jerome jerome 4096 nov 18 17:06 .
drwxr-xr-x  3 root   root   4096 nov 18 16:58 ..
-rw-r--r--  1 jerome jerome  220 mar 31  2024 .bash_logout
-rw-r--r--  1 jerome jerome 3771 mar 31  2024 .bashrc
drwxrwxr-x  6 jerome jerome 4096 nov 18 17:05 .cache
drwxrwxr-x  6 jerome jerome 4096 nov 18 17:05 .config
drwxr-xr-x  2 jerome jerome 4096 nov 18 17:04 Desktop
drwxr-xr-x  2 jerome jerome 4096 nov 18 17:04 Documents
drwxr-xr-x  2 jerome jerome 4096 nov 18 17:04 Downloads
drwx----- 4 jerome jerome 4096 nov 18 17:04 .local
drwxr-xr-x  2 jerome jerome 4096 nov 18 17:04 Music
-rw-rw-r--  1 jerome jerome    0 nov 18 17:06 myfile.txt
drwxr-xr-x  2 jerome jerome 4096 nov 18 17:04 Pictures
-rw-r--r--  1 jerome jerome  807 mar 31  2024 .profile
drwxr-xr-x  2 jerome jerome 4096 nov 18 17:04 Public
-rw-r--r--  1 jerome jerome    0 nov 18 17:05 .sudo_as_admin_successful
drwxr-xr-x  2 jerome jerome 4096 nov 18 17:04 Templates
-rw-----  1 jerome jerome 4732 nov 18 17:05 .xsession-errors
jerome@jerome-virtualbox:~$
```

3.5) DISK AND FILES MANAGEMENT

```
jerome@jerome-virtualbox:~$ mkdir c-code
jerome@jerome-virtualbox:~$ cd c-code
jerome@jerome-virtualbox:~/c-code$ ls -la
total 8
drwxrwxr-x  2 jerome jerome 4096 nov 18 17:08 .
drwxr-x--- 14 jerome jerome 4096 nov 18 17:08 ..
jerome@jerome-virtualbox:~/c-code$ cd ..
jerome@jerome-virtualbox:~$ ls -l
total 36
drwxrwxr-x 2 jerome jerome 4096 nov 18 17:08 c-code
drwxrwxr-x 2 jerome jerome 4096 nov 18 17:04 Desktop
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Documents
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Downloads
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Music
-rw-rw-r-- 1 jerome jerome  0 nov 18 17:06 myfile.txt
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Pictures
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Public
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Templates
drwxr-xr-x 2 jerome jerome 4096 nov 18 17:04 Videos
jerome@jerome-virtualbox:~$
```

3.5) DISK AND FILES MANAGEMENT

3.5) DISK AND FILES MANAGEMENT

\$ vim file.txt ### I -> insert, ESC -> exit insert, :wq -> write file and quit

\$ nano file.txt

\$ cat myfile.txt

\$ head myfile.txt

\$ tail myfile.txt

\$ more myfile.txt

\$ grep mystring myfile.txt

\$ chown owner:group name

\$ chgrp groupname filename.txt

\$ wc -l myfile.txt

3.5) DISK AND FILES MANAGEMENT

\$ ln -s /home theHomeDirectory ###symbolic link

\$ ls -la theHomeDirectory

\$ rm theHomeDirectory

\$ tree

3.5) DISK AND FILES MANAGEMENT

\$ mount

\$ umount

\$ fdisk ### **dangerous** command: changes the disk partitions!

\$ df

\$ du directoryname/

3.6) MEMORY

\$ free ### free -g ### free -h

```
jerome@jerome-virtualbox:~$ free
              total        used        free      shared  buff/cache   available
Mem:      4010564       785496      2724436        11032        732980      3225068
Swap:      524284           0         524284
jerome@jerome-virtualbox:~$
```

```
jerome@jerome-virtualbox:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:      3,8Gi         767Mi      2,6Gi        10Mi        718Mi      3,1Gi
Swap:      511Mi           0B         511Mi
jerome@jerome-virtualbox:~$ █
```

3.6) MEMORY

\$ vmstat ### vmstat 1 3

```
jerome@jerome-virtualbox:~$ vmstat 1 3
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
r  b   swpd   free   buff   cache   si   so    bi    bo    in    cs  us  sy  id  wa  st  gu
1  0       0 2724436 56100 676924    0    0    473    77   439    2   1   2  97   0   0   0
0  0       0 2724436 56100 676964    0    0     0     0   483   230  0   0  100   0   0   0
0  0       0 2724436 56100 676964    0    0     0     0   224   183  1   1  98   0   0   0
jerome@jerome-virtualbox:~$
```

jerome@jerome-virtualbox:~\$ man vmstat

```
VMSTAT(8)
System Administration
NAME
    vmstat - Report virtual memory statistics
SYNOPSIS
    vmstat [options] [delay [count]]
DESCRIPTION
    vmstat reports information about processes, memory, paging, block IO, traps, disks and cpu activity.
    The first report produced gives averages since the last reboot. Additional reports give information on a sam-
    pling period of length delay. The process and memory reports are instantaneous in either case.
OPTIONS
    delay The delay between updates in seconds. If no delay is specified, only one report is printed with the av-
    erage values since boot.
    count Number of updates. In absence of count, when delay is defined, default is infinite.
    -a, --active
        Display active and inactive memory, given a 2.5.41 kernel or better
```

3.6) MEMORY

\$ top

```

jerome@jerome-virtualbox: ~
jerome@jerome-virtualbox: ~ 93x25
top - 12:58:37 up 27 min, 1 user, load average: 0,04, 0,05, 0,09
Tasks: 170 total, 1 running, 169 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0,7 us, 2,5 sy, 0,0 ni, 96,5 id, 0,0 wa, 0,0 hi, 0,4 si, 0,0 st
MiB Mem : 3916,6 total, 2657,4 free, 767,6 used, 718,5 buff/cache
MiB Swap: 512,0 total, 512,0 free, 0,0 used. 3149,0 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    860 root        20   0  321316  87464  61656 S   6,6   2,2   0:21.19 Xorg
   1183 jerome      20   0 1099928 162424 100420 S   2,7   4,0   0:04.98 pcmanfm-qt
   1109 jerome      20   0  221596   3300   2900 S   0,7   0,1   0:05.06 VBoxClient
   1185 jerome      20   0  475140 105836  88440 S   0,7   2,6   0:00.68 lxqt-notificati
   1740 jerome      20   0  586572  77388  51196 S   0,7   1,9   0:02.55 terminator
    65 root        20   0      0      0      0 I   0,3   0,0   0:00.22 kworker/u9:3-events_+
   1142 jerome      20   0  151640  23492  18956 S   0,3   0,6   0:00.60 openbox
   1188 jerome      20   0  293072  57240  43672 S   0,3   1,4   0:01.07 lxqt-runner
   1190 jerome      20   0    4644   3348   3036 S   0,3   0,1   0:00.74 xscreensaver
   1320 jerome      20   0    515136  54680  45132 S   0,3   1,4   0:00.80 lxqt-powermanag
   1903 jerome      20   0    17388   6040   3864 R   0,3   0,2   0:00.02 top
    1 root        20   0    22436  13408   9440 S   0,0   0,3   0:02.56 systemd
    2 root        20   0      0      0      0 S   0,0   0,3   0:00.02 kthreadd
    3 root        20   0      0      0      0 S   0,0   0,0   0:00.00 pool_workqueue_relea+
    4 root        20   0      0      0      0 S   0,0   0,0   0:00.00 kworker/R-rcu_gp
    5 root        20   0      0      0      0 I   0,0   0,0   0:00.00 kworker/R-sync_wq
    6 root        20   0      0      0      0 I   0,0   0,0   0:00.00 kworker/R-kvfree_rcu+
    7 root        20   0      0      0      0 I   0,0   0,0   0:00.00 kworker/R-slub_flush+

```

3.6) MEMORY

\$ atop

jerome@jerome-virtualbox: ~

jerome@jerome-virtualbox: ~ 93x25

ATOP - jerome-virtualbox 2025/11/19 13:00:01 29m3s elapsed

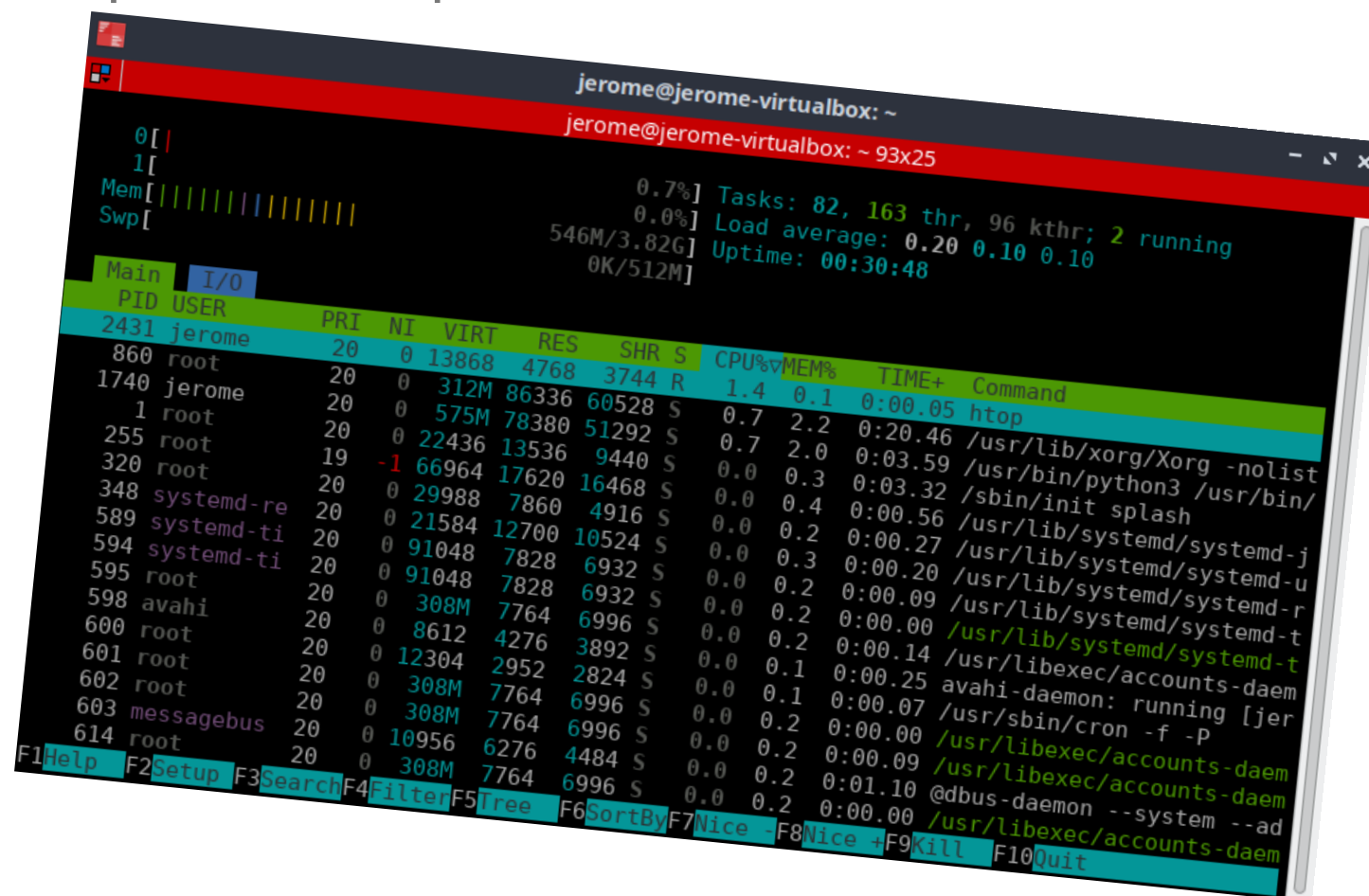
PRC	sys	52.99s	user	17.77s	#proc	177	#tidle	64	#exit	0
CPU	sys	3%	user	2%	irq	1%	idle	194%	wait	0%
cpu	sys	1%	user	1%	irq	1%	idle	96%	cpu000 w	0%
cpu	sys	1%	user	0%	irq	0%	idle	98%	cpu001 w	0%
CPL	numcpu	2	avg1	0.08	avg5	0.05	csw	673065	intr	793518
MEM	tot	3.8G	free	2.6G	avail	3.1G	cache	639.8M	slab	103.1M
MEM	numnode	1	shmem	9.7M	shrss	0.0M	shswp	0.0M	pgtab	12.2M
SWP	tot	512.0M	free	512.0M	swcac	0.0M	vmcom	1.9G	vmlim	2.4G
PAG	migrate	0	pgin	170877	pgout	31296	swout	0	oomkill	0
PSI	cpusome	2%	memsome	0%	memfull	0%	iosome	0%	iofull	0%
DSK	sda		busy	2%	read	19307	write	3026	avio	1.09 ms
NET	transport		tcpi	228	tcpo	221	udpi	288	udpo	288
NET	network		ipi	542	ipo	567	ipfrw	0	deliv	539
NET	enp0s3	0%	pcki	866	pcko	351	si	4 Kbps	so	0 Kbps
NET	lo	----	pcki	239	pcko	239	si	0 Kbps	so	0 Kbps

PID	SYSCPU	USRCPU	RDELAY	BDELAY	VGROW	RGROW	ST	EXC	THR	S	CPUNR	CPU	CMD
860	16.79s	5.18s	7.66s	0.00s	312.7M	84.3M	N-	-	2	S	1	1%	Xorg
1109	3.89s	1.81s	5.84s	0.00s	216.4M	3.2M	N-	-	5	S	0	0%	VBoxClient
1183	3.11s	2.08s	5.35s	0.00s	1.0G	158.6M	N-	-	6	S	1	0%	pcmanfm-qt
1186	2.60s	0.82s	5.63s	0.00s	840.7M	124.5M	N-	-	5	S	1	0%	lxqt-panel
1	2.46s	0.86s	5.23s	0.00s	21.9M	13.2M	N-	-	1	S	0	0%	systemd
1740	1.35s	1.48s	1.03s	0.00s	572.8M	75.6M	N-	-	6	S	1	0%	terminator
10	2.53s	0.00s	1.68s	0.00s	0B	0B	N-	-	1	I	0	0%	kworker/0:1-cg

1/26

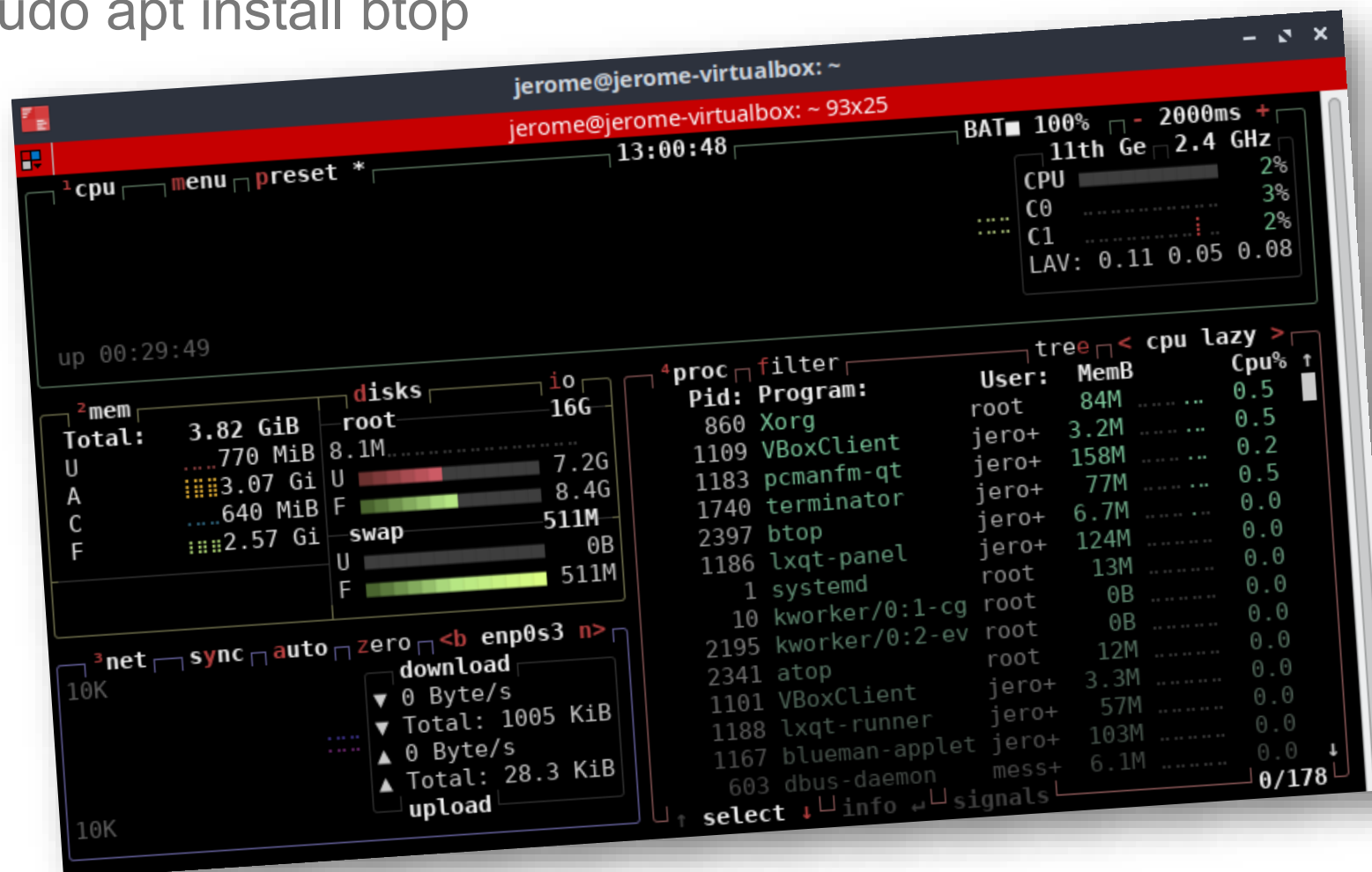
3.6) MEMORY

\$ htop #### sudo apt install htop



3.6) MEMORY

\$ btop ### sudo apt install btop



3.7) TASKS

\$ ps ### ps u ### ps a ### ps x ### ps aux ### ps -eLf

\$ kill -9 1234 ### kill -l : list the signals

\$ pgrep firefox ### finds all processes containing “firefox”

\$ ps aux | grep firefox ### finds all processes containing “firefox”

\$ killall firefox ### kills all the processes named “firefox”

3.8) NETWORK

\$ ifconfig

```
jerome@jerome-virtualbox:~$ ifconfig
Command 'ifconfig' not found, but can be installed with:
sudo apt install net-tools
jerome@jerome-virtualbox:~$ sudo apt install net-tools
```

```
jerome@jerome-virtualbox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:e003:80d5:acea:1869 prefixlen 64 scopeid 0x0<global>
    inet6 fd17:625c:f037:2:35de:c606:90cc:b33c prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:7a:c2:ee txqueuelen 1000 (Ethernet)
    RX packets 252 bytes 219541 (219.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 188 bytes 18034 (18.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 201 bytes 15904 (15.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 201 bytes 15904 (15.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

jerome@jerome-virtualbox:~$
```

3.8) NETWORK

\$ ping www.google.fr

```
jerome@jerome-virtualbox:~$ ping www.google.fr
PING www.google.fr (142.250.74.3) 56(84) bytes of data.
64 bytes from arn09s21-in-f3.1e100.net (142.250.74.3): icmp_seq=1 ttl=255 time=12.3 ms
64 bytes from arn09s21-in-f3.1e100.net (142.250.74.3): icmp_seq=2 ttl=255 time=12.6 ms
64 bytes from arn09s21-in-f3.1e100.net (142.250.74.3): icmp_seq=3 ttl=255 time=12.8 ms
^C
--- www.google.fr ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2008ms
rtt min/avg/max/mdev = 12.277/12.549/12.804/0.215 ms
jerome@jerome-virtualbox:~$
```

\$ netstat ### netstat -a

```
jerome@jerome-virtualbox:~$ netstat -a
Active Internet connections (servers and established)

```

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	_localdnsstub:domain	0.0.0.0:*	LISTEN
tcp	0	0	_localdnsproxy:domain	0.0.0.0:*	LISTEN
tcp	0	0	localhost:ipp	0.0.0.0:*	LISTEN
tcp	0	0	localhost:ipp	:::*	LISTEN
tcp6	0	0	_localdnsproxy:domain	0.0.0.0:*	
udp	0	0	_localdnsstub:domain	0.0.0.0:*	
udp	0	0	jerome-virtualbo:bootpc	_gateway:bootps	ESTABLISHED
udp	0	0	0.0.0.0:35115	0.0.0.0:*	
udp	0	0	0.0.0.0:mdns	0.0.0.0:*	
udp	0	0	:::34552	:::*	
udp6	0	0	:::mdns	:::*	
udp6	0	0	:::ipv6-icmp	:::*	7

```

Active UNIX domain sockets (servers and established)

```

Proto	RefCnt	Flags	Type	State	I-Node	Path
unix	3	[]	STREAM	CONNECTED	9448	
unix	3	[]	STREAM	CONNECTED	11998	
unix	3	[]	STREAM	CONNECTED	11909	
unix	3	[]	STREAM	CONNECTED	10216	@/tmp/.X11-unix/X0
unix	3	[]	STREAM	CONNECTED	11606	/run/dbus/system_bus_socket

3.8) NETWORK

```
$ tcpdump -i wifi0    ### sudo apt install tcpdump
```

3.9) PATTERNS AND REGULAR EXPRESSIONS

- * replaces any number of characters
- ? replaces exactly one character
- . replaces any one character except newline
- ^ start of the string
- \$ end of the string

\$ ls *.png #### list all the .png files

\$ ls *img* #### list all files and directories having img somewhere in their name

\$ ls alphabet-?.txt #### list all alphabet-[one letter].txt files

\$ ls -l | grep ^H #### everything starting with a H

3.10) PIPES AND REDIRECTIONS

- You can redirect results of commands using `>` and `>>`

`$ ls -l > mylist.txt` #### list all the files and folders in the text file mylist.txt (the content is deleted)

`$ ls -l >> my list.txt` #### list all the files and folders at the end of the text file mylist.txt (the content is added after the existing one)

- You can chain commands using pipes `|` : the output of one command is given as input to another one:

`$ ps aux | grep sh`

`$ ps aux | wc -l`

`$ ps aux | grep firefox | wc -l` #### how many firefox processes are running

3.11) SYSTEM INFORMATION

\$ which qterminal

\$ dmesg ### displays system information from the start of the system

\$ env ### print the environment variables of the system

\$ jobs ### print the jobs running inside the terminal

\$ uname ###-a or cat /proc/version

\$ cat /proc/cpuinfo

\$ cat /proc/meminfo

3.12) PACKAGES MANAGEMENT

\$ sudo apt update

```
jerome@jerome-virtualbox:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
135 packages can be upgraded. Run 'apt list --upgradable' to see them.
jerome@jerome-virtualbox:~$
```


3.12) PACKAGES MANAGEMENT

\$ sudo apt upgrade

```
jerome@jerome-virtualbox:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  vlc-plugin-qt libvlc5 libmagickcore-6.q16-7t64 libzvbi-common vlc-data
  libvlccore9 imagemagick vlc-bin libmagickcore-6.q16-7-extra libcjson1
  libpostproc57 vlc-plugin-samba libavcodec60 libzvbi0t64 vlc-plugin-notify
  libavutil58 imagemagick-6.q16 libswscale7 vlc-plugin-access-extra
  vlc-plugin-skins2 vlc-plugin-svg vlc-plugin-video-splitter libswresample4
  imagemagick-6-common vlc-plugin-video-output 7zip libavformat60 libvlc-bin
  vlc-plugin-base libavfilter9 libmagickwand-6.q16-7t64
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following NEW packages will be installed:
  libllvm20 linux-headers-6.14.0-35-generic linux-hwe-6.14.0-35
  linux-image-6.14.0-35-generic linux-modules-6.14.0-35-generic
  linux-tools-6.14.0-35-generic
The following packages will be upgraded:
  bind9-dnsutils bind9-host bind9-libs bluez bluez-cups-cups-core-cups-drivers cups-daemon cups-in
  cups-libs cups-modules-extra-6.14.0-35-generic xserver-xorg-video-nouveau xserver-xorg-video-vesa xxd
135 upgraded, 8 newly installed, 0 to remove and 0 not upgraded.
Need to get 907 MB of archives.
After this operation, 443 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

3.12) PACKAGES MANAGEMENT

\$ sudo apt upgrade

Several minutes later...

3.12) PACKAGES MANAGEMENT

```
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Setting up libgtk-4-1:amd64 (4.14.5+ds-0ubuntu0.5) ...
Setting up libgtk-4-bin (4.14.5+ds-0ubuntu0.5) ...
Setting up libgtk-4-media-gstreamer (4.14.5+ds-0ubuntu0.5) ...
Processing triggers for linux-image-6.14.0-35-generic (6.14.0-35.35~24.04.1) ...
/etc/kernel/postinst.d/initramfs-tools:
update-initramfs: Generating /boot/initrd.img-6.14.0-35-generic
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file `/etc/default/grub'
Sourcing file `/etc/default/grub.d/lubuntu-grub-theme.cfg'
Generating grub configuration file ...
Found theme: /usr/share/grub/themes/lubuntu-grub-theme/theme.txt
Found linux image: /boot/vmlinuz-6.14.0-35-generic
Found initrd image: /boot/initrd.img-6.14.0-35-generic
Found linux image: /boot/vmlinuz-6.14.0-27-generic
Found initrd image: /boot/initrd.img-6.14.0-27-generic
Found memtest86+x64 image: /boot/memtest86+x64.bin
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
jerome@jerome-virtualbox:~$
```

3.12) PACKAGES MANAGEMENT

\$ sudo apt search terminator

```
jerome@jerome-virtualbox:~$ sudo apt search terminator
Sorting... Done
Full Text Search... Done
aseqjoy/noble 0.0.2-1build3 amd64
  Joystick to ALSA MIDI Sequencer Converter
libfile-loadlines-perl/noble,noble 1.046-2 all
  Load lines from files and network
libproc-terminator-perl/noble,noble 0.5-3 all
  module to conveniently terminate processes
node-has-flag/noble,noble 4.0.0-3 all
  check if argv has a specific flag
no-page-terminator-16/noble 0.1.7-9 amd64
```

```
terminator/noble,noble,now 2.1.3-1 all [installed]
  multiple GNOME terminals in one window
terminatorx/noble 4.2.0-1build2 amd64
  realtime audio synthesizer
transtermhp/noble 2.09-5 amd64
  find rho-independent transcription terminators in bacterial genomes
jerome@jerome-virtualbox:~$
```

\$ sudo apt install terminator

```
jerome@jerome-virtualbox:~$ sudo apt install terminator
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  terminator
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 356 kB of archives.
After this operation, 2 275 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 terminator all 2.1.3-1 [356 kB]
Fetched 356 kB in 1s (516 kB/s)
Selecting previously unselected package terminator.
(Reading database ... 303421 files and directories currently installed.)
Preparing to unpack .../terminator_2.1.3-1_all.deb ...
Unpacking terminator (2.1.3-1) ...
Setting up terminator (2.1.3-1) ...
```

```
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
jerome@jerome-virtualbox:~$ █
```

3.13) OTHER COMMANDS

\$ `cal` ### prints the calendar, “`sudo apt install ncal`”

```
jerome@jerome-virtualbox:~$ cal
November 2025
sö må ti on to fr lö
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30
jerome@jerome-virtualbox:~$
```

\$ `bc` ### GNU calculator on the command line

```
jerome@jerome-virtualbox:~$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
123*3.14
386.22
-2*56
-112
-1*-1
1
quit
jerome@jerome-virtualbox:~$
```


3.13) OTHER COMMANDS

\$ env #### print the environment variables of the system

```
jerome@jerome-virtualbox:~$ env
SHELL=/bin/bash
WINDOWID=0
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-Lubuntu:/etc/xdg:/etc:/usr/share
XDG_SESSION_PATH=/org/freedesktop/DisplayManager/Session1
XDG_MENU_PREFIX=lxqt-
SAL_VCL_QT5_USE_CAIRO=true
LANGUAGE=
LC_ADDRESS=sv_SE.UTF-8
LC_NAME=sv_SE.UTF-8
SSH_AUTH_SOCK=/tmp/ssh-kWfg0AYSImwS/agent.998
XDG_DATA_HOME=/home/jerome/.local/share
XDG_CONFIG_HOME=/home/jerome/.config
DESKTOP_SESSION=Lubuntu
LXQT_SESSION_CONFIG=session
LC_MONETARY=sv_SE.UTF-8
SSH_AGENT_PID=1052
XDG_SEAT=seat0
PWD=/home/jerome
XDG_SESSION_DESKTOP=LXQt
LOGNAME=jerome
QT_QPA_PLATFORMTHEME=lxqt
XDG_SESSION_TYPE=x11

USER=jerome
COLORFGBG=15;0
DISPLAY=:0
SHLVL=1
LC_TELEPHONE=sv_SE.UTF-8
LC_MEASUREMENT=sv_SE.UTF-8
XDG_VTNR=2
XDG_SESSION_ID=3
XDG_RUNTIME_DIR=/run/user/1000
QT_PLATFORM_PLUGIN=lxqt
LC_TIME=sv_SE.UTF-8
XDG_DATA_DIRS=/usr/share/Lubuntu:/usr/local/share:/usr/share:/usr/share
BROWSER=firefox
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
SAL_USE_VCLPLUGIN=qt5
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
LC_NUMERIC=sv_SE.UTF-8
_=usr/bin/env
OLDPWD=/home/jerome/c-code
jerome@jerome-virtualbox:~$
```

3) COMMANDS

- Demonstration of different commands on Linux filesystem...
- **RECALL:** You need to come to the lectures to get the information

ANY QUESTIONS?